DynamiClinic

From a scoping review of 81 multi-agent studies to a system that switches its architecture per clinical case.

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github.com/LaansDole/multi-agent-simulation
branch threejs-devall

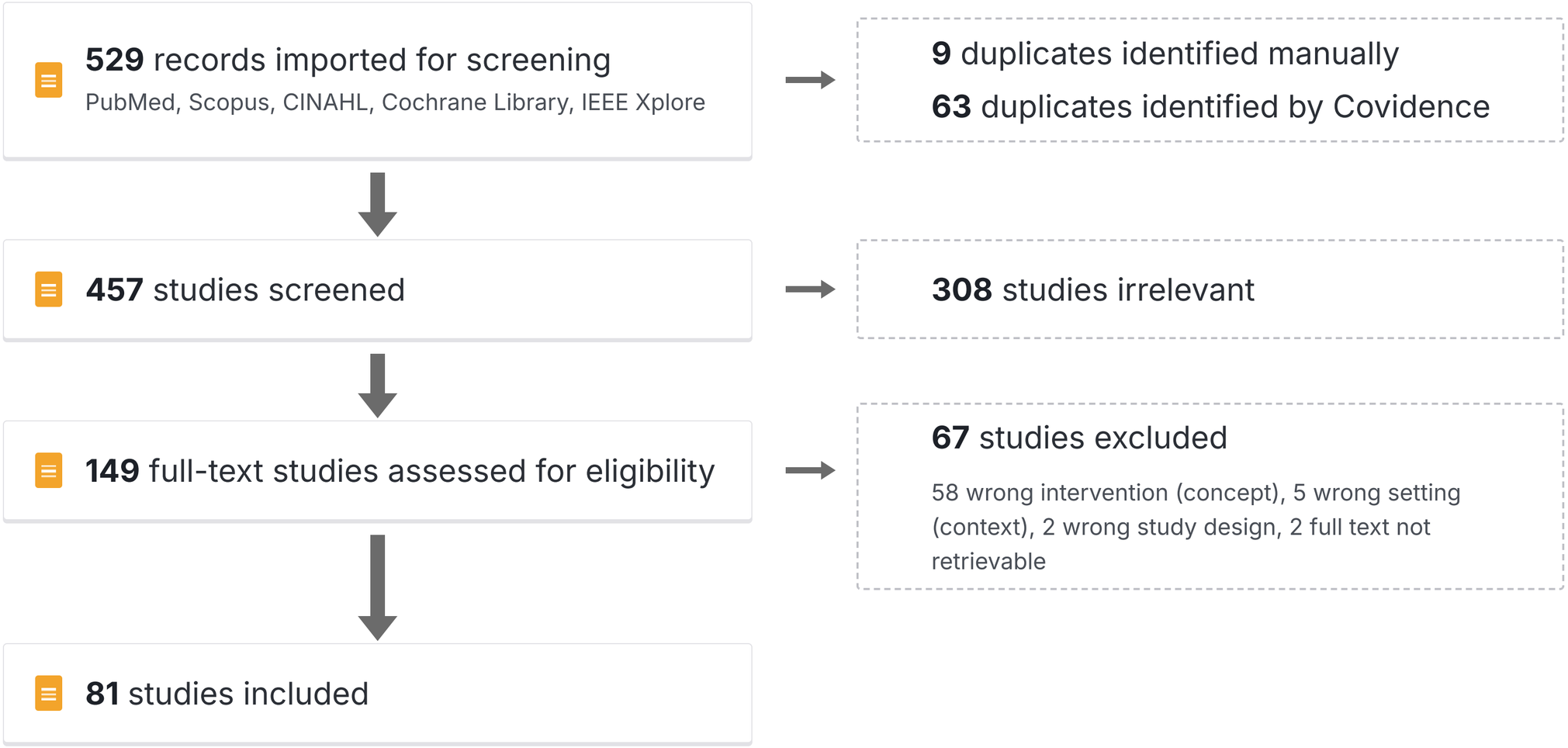
How do teams of LLM agents work in healthcare?

Scope

- Two or more LLM agents on one healthcare task
- One agent revises its output in response to another
- Five questions: where, architecture, models, evaluation, cost

Method

- 5 databases, June 2026, records 2022-2026
- Arksey and O'Malley, reported per PRISMA-ScR
- Screened in Covidence



PRISMA-ScR flow, Fig. 1 of the scoping review

From the review's themes to one adaptive system

REVIEW THEMES

Hierarchical coordination
30 studies

Debate and consensus
35 studies

Dynamic team assembly
11 studies, chosen once by difficulty

→

GAP

Systems pick a pattern up front; none switches between hierarchy, consensus and debate **as the case unfolds**.

Collaboration takes 4-12x longer; comparison with a single model is not routine.

→

PROPOSAL

Hierarchical mode

Collaborative mode

Adversarial mode

Meta-Controller switches per case

Themes: scoping review, sections 3 and 4. Gap as identified for the DynaClinic proposal.

A Meta-Controller makes the topology a variable

Two scores, one rule

Complexity C: how dense the case is (comorbidities, drugs, conflicting labs).
Uncertainty U: entropy of a generalist's first differential.

```
U > tau_u → adversarial
C ⩽ tau_c → hierarchical
otherwise → collaborative
```

A Conflict Detector escalates collaborative to adversarial when the specialists diverge. tau_u is calibrated per model (gpt-6-luna: 0.15 multiple choice, 0.59 open-ended).

Hierarchical
SPEED

Primary Care Agent as the hub; specialists only on request.

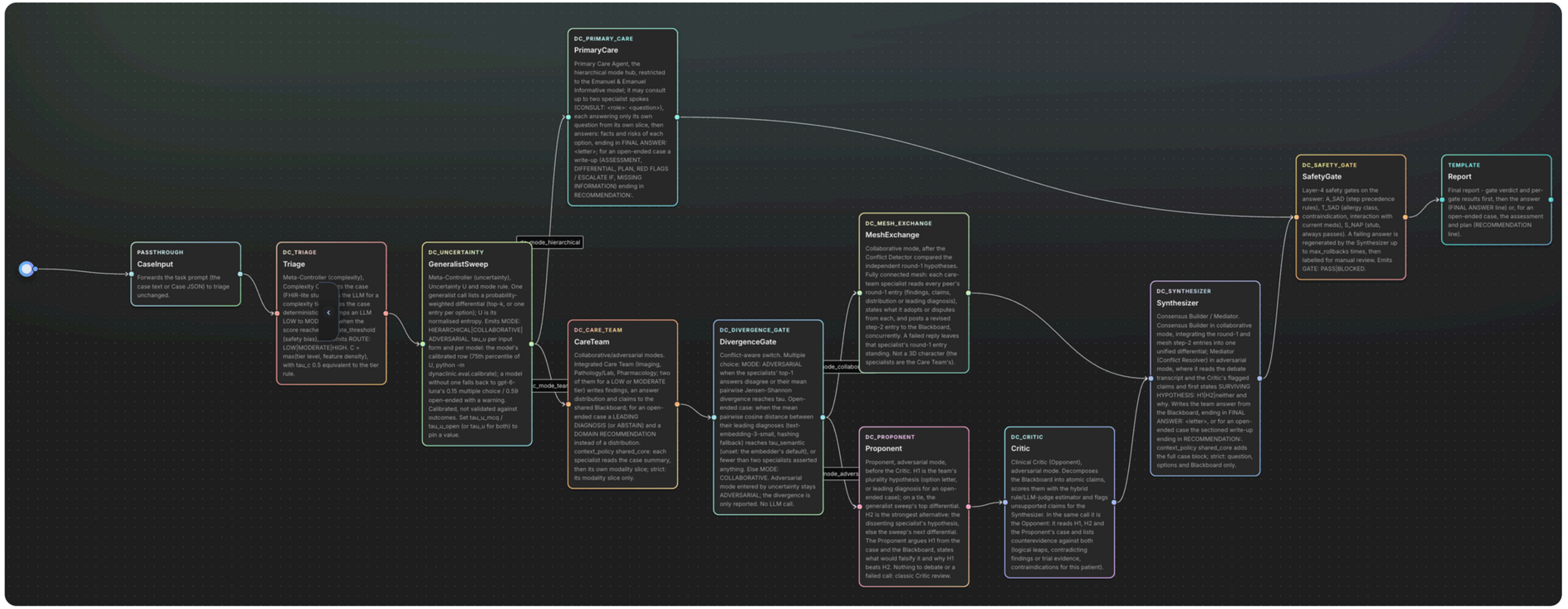
Collaborative
BREADTH

Imaging, Pathology/Lab, Pharmacology on a shared Blackboard; mesh exchange; Consensus Builder.

Adversarial
DEPTH

Proponent defends H1, Opponent (Clinical Critic) attacks, Mediator decides.

Built as one workflow on the DevAll platform



Hierarchical
`GeneralistSweep → PrimaryCare`

Collaborative
`CareTeam → DivergenceGate → MeshExchange`

Adversarial
`DivergenceGate → Proponent → Critic`

Rendered by DevAll's workflow editor from yaml_instance/dynaclinic_mvp.yaml. All paths end in Synthesizer, SafetyGate, Report.

Four pieces behind the graph

10

Custom node types

Every DynaClinic step in the graph is a new DevAll node written in Python: triage, uncertainty sweep, primary care, care team, divergence gate, mesh, proponent, critic, synthesizer, safety gate. One YAML file wires them.

3

Evaluation arms

Single: one model call answers. **Static team**: the full team and critic every time. **Dynamic**: DynamiClinic picks the mode. Same questions, same model.

3D

Office view

Imports any DevAll workflow; each agent is a character. On a hand-off the agents meet and talk, and each nameplate shows the agent's current role.

1,030

Automated tests

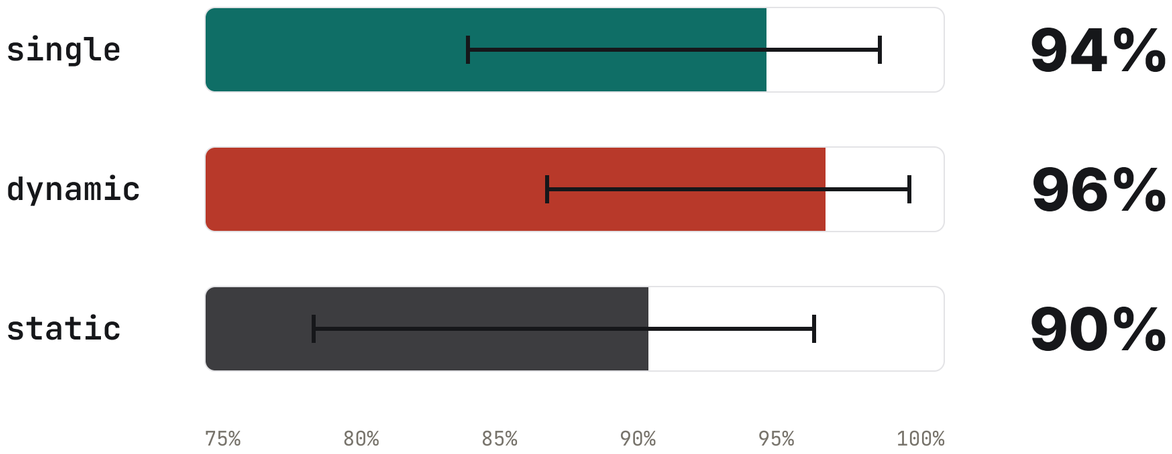
Run offline with a fake model, no API calls: mode rule, thresholds, safety gates, engine fixes. Real runs are kept separate.

dynaclinic/ (Python), yaml_instance/dynaclinic_mvp.yaml, the-delegation/ (React + Three.js). Stubs: FHIR/Neo4j, MCTS planner, BioBERT/Qdrant memory.

Works end to end; not yet better than one agent

MedQA-USMLE, 50 test questions, gpt-6-luna

Accuracy with 95% interval, one run per arm.



	SINGLE	DYNAMIC	STATIC
gpt-6-luna calls / question	1.00	6.94	6.00
gpt-6-luna mean latency	5.3 s	32.0 s	29.9 s
qwen3.5-4b accuracy (local)	68%	60%	62%

The team's answer follows the Meta-Controller's **first look** (gpt-6-luna 47 of 50, qwen 31 of 50); on exam questions the specialists add little. All intervals overlap: exactly the **single-model comparison** the review asks for.

MedQA: Jin et al. 2021, 4-option test split, seed 2026 sample. qwen: dynamic with its own calibrated tau_u; single vs dynamic p = 0.34.

Next: make the team earn its cost

1 **Harder cases, mechanism by mechanism**

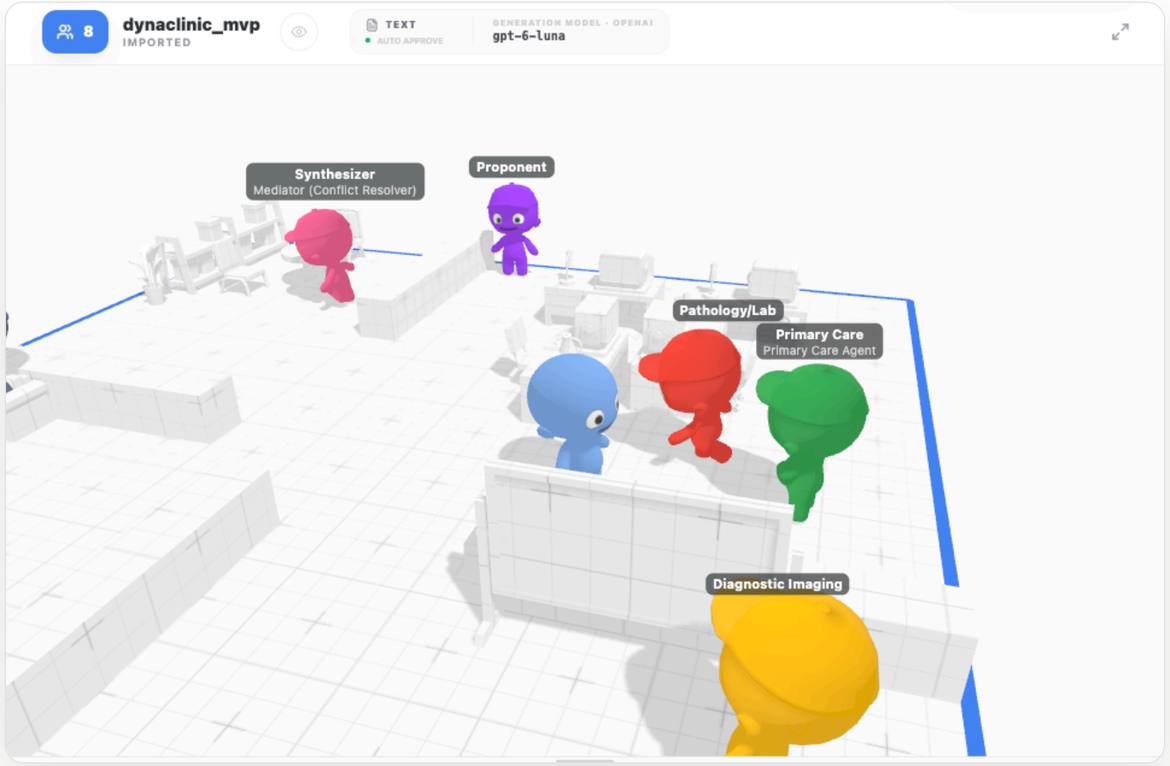
A benchmark where one agent fails (MedXpertQA); ablate mesh, debate and gates one at a time.

2 **A better controller**

Validate tau against outcomes; give the first look a vote; a Mediator that weighs evidence, not plurality.

3 **Toward clinical validation**

Clinician review of debate transcripts, then the prospective studies the review found missing.



LIVE DEMO: ONE CASE PER MODE

A	Cellulitis, sulfonamide allergy	HIERARCHICAL
B	Renal colic on phenelzine	COLLABORATIVE
C	30 weeks pregnant, chest pain	ADVERSARIAL